

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, yellow). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, emphasizing the text overlay.

C#

Level 1

**Managed Vs Unmanaged
Code**



C#

Level 1

Managed Vs Unmanaged Code

Definition

Managed Code

-  Code that executes under the control of the CLR.

-  CLR provides services such as GC, exception handling, type safety, and runtime support.

Unmanaged Code

-  Code that executes outside the CLR.

-  Commonly associated with native C/C++ and operating-system APIs.

Explanation

Managed Execution

-  C# is normally compiled into IL.
-  CLR loads the assembly and manages its execution.
-  Garbage Collector manages ordinary managed objects.

Unmanaged Execution

-  Native code is compiled directly into machine code.
-  It executes outside the CLR.
-  Resource management is handled by the programmer or native runtime mechanisms.

Procedures

1. Managed: $C\# \rightarrow IL \rightarrow CLR \rightarrow JIT \rightarrow \text{Native Code}$.
2. Unmanaged: $\text{Native Source} \rightarrow \text{Native Compiler} \rightarrow \text{Native Code}$.
3. Managed code can communicate with native code through interoperability mechanisms.
4. P/Invoke is one example of managed-to-native interoperability.

💡 Example

▨ Managed:

$C\# \rightarrow IL \rightarrow CLR \rightarrow JIT \rightarrow \text{Native Code}$

▤ Unmanaged:

$C++ \rightarrow \text{Native Code} \rightarrow \text{Operating System}$

➡ Result

Managed code runs under the CLR, while unmanaged code runs outside the CLR.

Note

- ◆ C++ can be managed or unmanaged depending on how it is compiled.
- ◆ Managed memory is handled by the .NET GC.
- ◆ Unmanaged resources may require explicit cleanup.

⊘ Warning

- ◆ Do not memorize "C# = managed, C++ = unmanaged" as an absolute rule.
- ◆ C++/CLI can interact with managed .NET code.



Comparison

▨ Managed Code

- ✓ Runs under the CLR.
- ✓ Uses .NET runtime services.
- ✓ Garbage Collection manages managed memory.
- ✓ Common in C# applications.

▨ Unmanaged Code

- ✓ Runs outside the CLR.
- ✓ Common in native C/C++.
- ✓ Provides more direct native/system interaction.
- ✓ Resource management may require explicit handling.

✦✦ Summary

- ✓ Managed code runs under the CLR.
- ✓ Unmanaged code runs outside the CLR.
- ✓ CLR provides GC and other runtime services.
- ✓ Native C/C++ is commonly unmanaged.
- ✓ Managed and unmanaged code can communicate through interoperability.
- ✓ The classification depends on how the code is executed.

Thank you 😊